Week #7: [Osteology & Joints of the Proximal Upper Extremity]

(Non-hyperlinked = Dr. Ken Faber Module)

- Introduction to Module

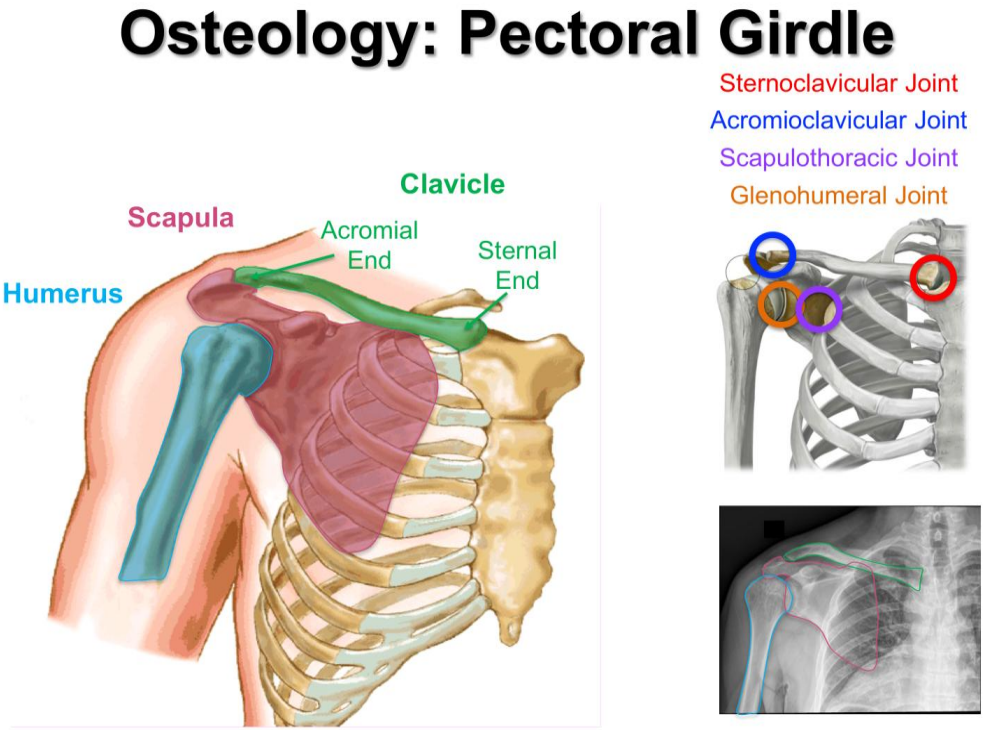
Osteology of the proximal Upper extremity

- Osteology (27 min)

Joints of the Proximal Upper Extremity

- Anatomy of the Joints of the Pectoral Girdle (30 min)
- Approach to Acromioclavicular Joint Problems (11 min)
- Approach to Glenohumeral Joint Problems (40 min)

Osteology



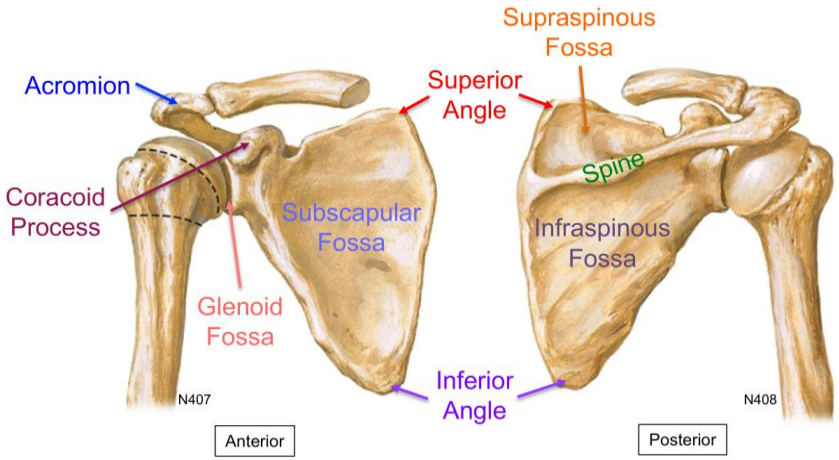
Osteology:

a. The pectoral girdle consists of the clavicle and the scapula. There are four joints that comprise the pectoral girdle
   i. **Sternoclavicular joint** – between the proximal clavicle and the manubrium of the sternum
   ii. **Acromioclavicular joint** - between the distal clavicle and the acromion process of the scapula
   iii. **Scapulothoracic joint** – between the deep aspect of the scapula and the thoracic wall
   iv. **Glenohumeral joint** – between the glenoid fossa of the scapula and the head of the humerus

Clavicle – Forms an S-shape with its medial half convex anteriorly and the lateral half concave anteriorly

a. The medial end that articulates with the manubrium is enlarged and triangular while the acromial end is more flat where it articulates with the acromion process

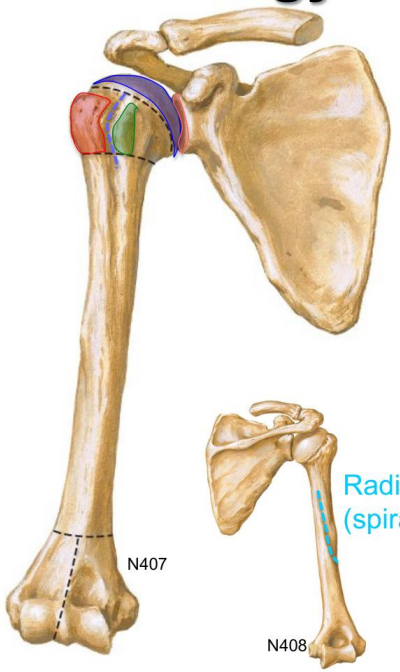
Osteology: Scapula



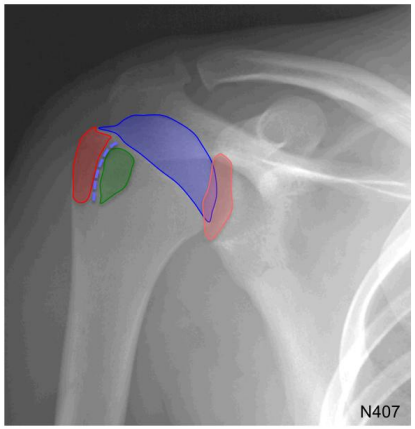
Scapula – triangular flat bone that is essential for overhead movements of the upper extremity

1. **Acromion Process** – articulates with the clavicle to form the acromioclavicular joint
2. **Coracoid Process** – large projection for muscle attachments
3. **Glenoid Fossa** – shallow depression that articulates with the head of the humerus to form the glenohumeral joint
    a. ***Supraglenoid Tubercle*** – projection at the superior aspect of the glenoid fossa for the origin of the long head of the biceps brachii
    b. ***Infraglenoid Tubercle*** – projection at the inferior aspect of the glenoid fossa for the origin of the long head of the triceps brachii
4. **Superior Angle and Inferior Angle** – sharp ridges for muscle attachments and landmarking
5. **Subscapular Fossa** – concavity in which the subscapularis muscle lies
6. **Spine of the scapular** – divides the posterior scapula into the supraspinous fossa and the infraspinous fossa
    a. The ***supraspinatus muscle*** lies within the supraspinous fossa
    b. The ***infraspinatus muscle*** lies within the infraspinous fossa
7. **Suprascapular Notch** – forms the suprascapular foramen when the transverse scapular ligament crosses it. Neurovascular structures use this notch to travel to and from the anterior aspect of the scapula

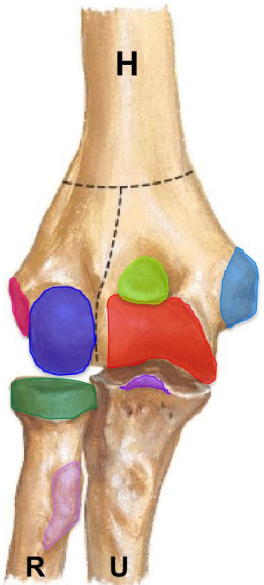
Osteology: Proximal Humerus



Head
Glenoid Fossa
Greater Tubercle
Lesser Tubercle
Intertubercular Groove
Radial Groove (spiral groove)

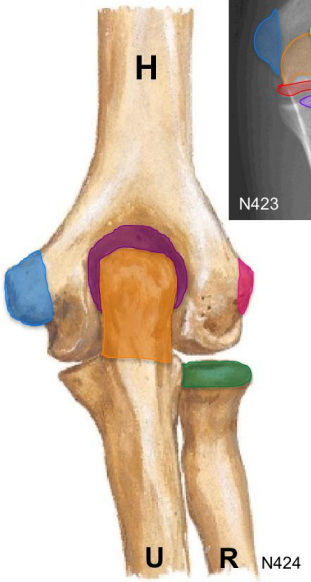


Osteology: Distal Humerus



Anterior View
Right

Trochlea
Capitulum
Medial Epicondyle
Lateral Epicondyle
Radial Head
Coronoid
Coronoid Fossa
Olecranon
Olecranon Fossa
Radial Tuberosity



Posterior View
Right



Humerus – only bone of the arm

1. **Head** – articulates with the scapula at the glenohumeral joint
2. **Greater tubercle** – prominence on the superolateral humerus; insertion site
3. **Lesser tubercle** – prominence on the superoanterior humerus; insertion site
4. **Intertubercular (bicipital) groove** – between greater and lesser tubercles; tendon of ***long head of biceps brachii*** runs through groove
5. **Anatomical neck** – separates head from tubercles
6. **Surgical neck** – narrowing distal to head and tubercles; common fracture site
7. **Radial (spiral) groove** – located on posterior shaft; ***radial nerve and profunda brachii artery*** run along groove
8. **Medial epicondyle** – medial protuberance at the distal end of the humerus; origin of the ***common flexor tendon for the forearm flexors***; ulnar nerve courses posterior to it
9. **Lateral epicondyle** – lateral protuberance at the distal end of the humerus; origin of the ***common extensor tendon for the forearm extensors***
10. **Capitulum** – lateral condyle; articulates with the head of the radius
11. **Trochlea** – medial condyle; articulates with the trochlear notch of the ulna
12. **Coronoid fossa** – depression superior to the trochlea anteriorly; articulates with the coronoid process during full flexion
13. **Olecranon fossa** – depression superior to the trochlea posteriorly; articulates with the olecranon process during full extension

Clinical Note: Fractures of the **surgical neck** of the humerus, often caused by low energy falls to the hand are especially common in elderly individuals with osteoporosis. This type of fracture may result in damage to the ***axillary nerve.*** Transverse fractures of the shaft of the humerus frequently result from a direct blow to the arm. This may result in injury to the ***radial nerve*** and/or ***deep brachial artery***. Fractures of the distal humerus may damage the **median nerve.** Fractures of the ***medial epicondyle*** may result in damage to the ***ulnar nerve.***

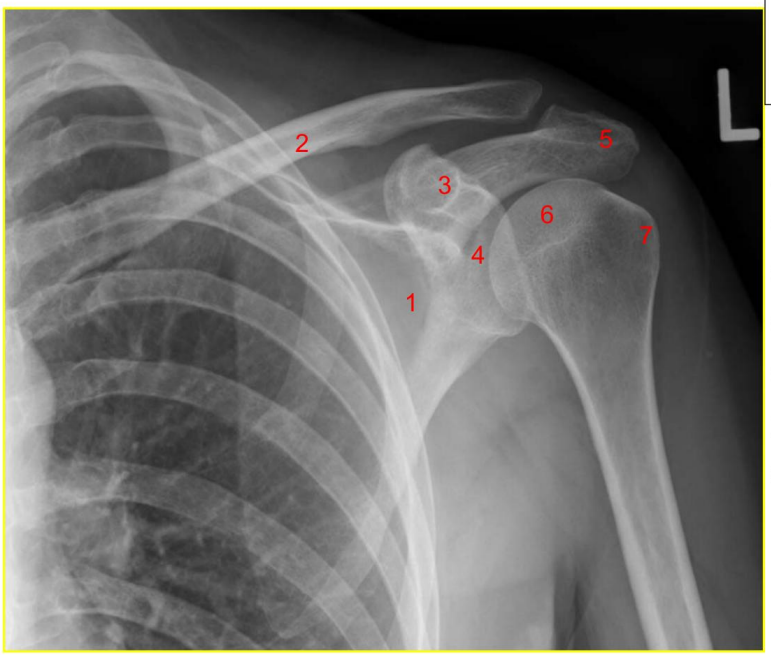
Radius – lateral and shorter bone of the forearm

1. **Head** – proximal portion that articulates with the capitulum of the humerus and radial notch of the ulna; pivots within the ***annular ligament***
2. **Neck** – narrow portion between the head and the radial tuberosity
3. **Radial tuberosity** - medial projection; insertion site for ***biceps brachii***

Ulna – medial and longer stabilizing bone of the forearm.

1. **Olecranon process** – bony eminence at posterior aspect of proximal ulna; insertion site for ***triceps brachii***
2. **Coronoid process** – triangular protuberance at anterior aspect of proximal ulna; insertion site of ***brachialis***
3. **Trochlear notch** – large depression between olecranon and coronoid process; articulates with trochlea
4. **Radial notch** – concavity on lateral side of coronoid process that articulates with the head of the radius

Knowledge Check



Shoulder internally rotated (AP view)

Acromion

Clavicle

Coracoid process

Glenoid fossa

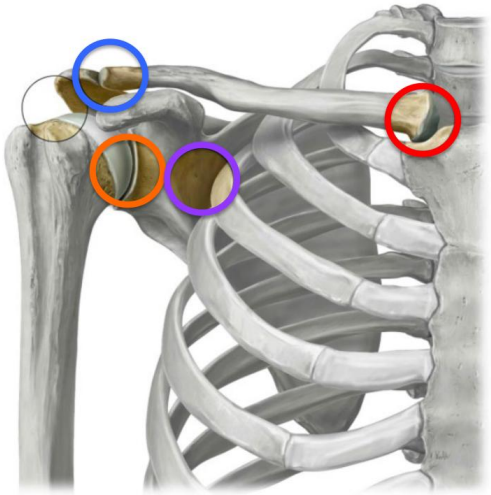
Greater tubercle

Head of humerus

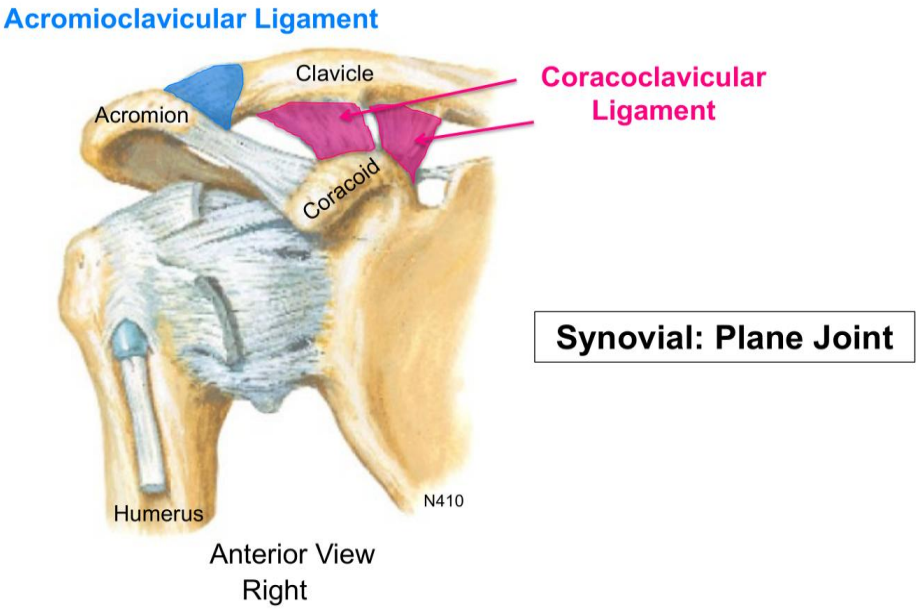
Scapula

Pectoral Girdle Joints

- 4 Joints
  1. Sternoclavicular
  2. Acromioclavicular
  3. Glenohumeral
  4. Scapulothoracic



Acromioclavicular (AC) Joint



Synovial: Plane Joint

Joints of the Pectoral Girdle

Scapulohumeral Rhythm: Full range of motion at the shoulder requires manipulation of the glenoid cavity position/angle; for muscles to function efficiently, their origins must be stable.

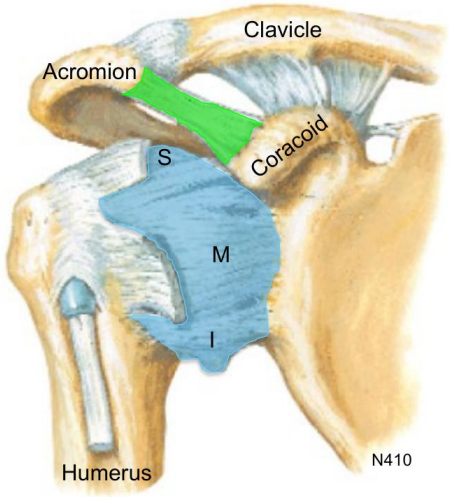
A. Sternoclavicular (SC) joint

- Only articulation conjoining the upper extremity to the axial skeleton.
- **Clinical Note:** Dislocations of the sternoclavicular joint are rare due to the strength of the surrounding ligaments. The clavicle is likely to fracture before the joint dislocates.

B. Acromioclavicular (AC) joint

1. ***Type:*** plane type of synovial joint; uniaxial
2. ***Articulation:*** acromial end of clavicle, acromion of scapula
   a. **Note:** joint space contains wedge-shaped articular disc
3. ***Ligaments:***
   a. Acromioclavicular ligament: strengthens joint superiorly
   b. Coracoclavicular ligament: pair of strong bands that anchor the clavicle to the coracoid process; confers great deal of stability/strength to the joint; enables the scapula and free limb to be passively suspended from the strut-like clavicle.
4. ***Movements:*** the acromion of the scapula rotates on the acromial end of the clavicle
5. **Clinical Note:** dislocation of the AC joint can result from direct or indirect force upon the joint. AC subluxations are usually associated with upward forces along the humerus and result in the rupture of the acromioclavicular ligament. AC dislocation is deemed more severe and is associated with blunt force trauma to the tip of the shoulder and results in the rupture of the strong, coracoclavicular ligament along with the acromioclavicular ligament; patient may present with clavicle displaced superiorly and acromion more prominent upon palpation.

Glenohumeral Joint



Anterior View
Right

Synovial: Ball & Socket Joint

Joint Capsule

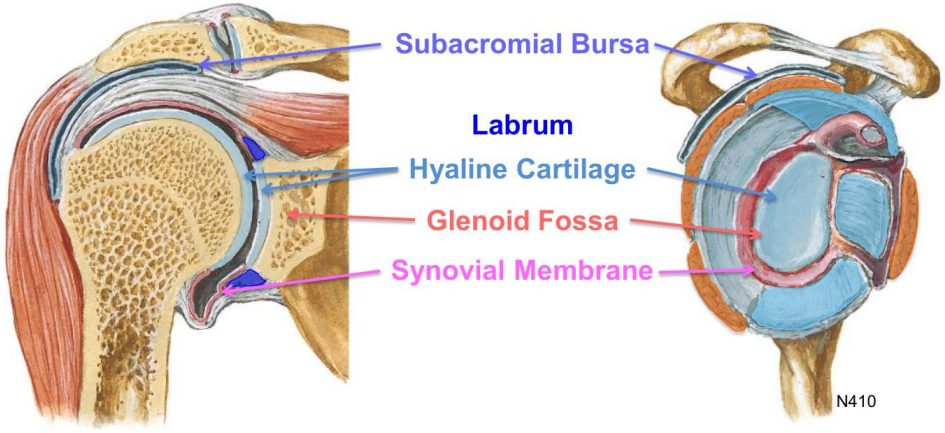
Glenohumeral Ligaments (3)

Coracoacromial Ligament

Glenohumeral Joint

Glenohumeral Ligaments (3)

Rotator Cuff Tendons



Coronal View
Right

Lateral View
Right

C. Glenohumeral (shoulder) joint

1. ***Type:*** **ball and socket** type of synovial joint; multiaxial
2. ***Articulation:*** large **humeral head;** shallow **glenoid fossa of scapula**
    a. **Mobility** is enhanced by (i) *poor articular congruency*; (ii) *several bursae;* (iii) *high degree of articular cartilage;* and (iv) *lax joint capsule.*
    b. **Stability i**s enhanced by the ring-like, fibrocartilaginous glenoid labrum (increases fit, deepens socket), the **musculotendinous rotator cuff**, and **ligament support.**
    c. The upper limb **range-of-motion** of the shoulder involves not only the **glenohumeral joint,** but also involves movement of the **SC and AC joints** and movement of the scapula on the thorax (conceptual '**scapulothoracic joint**').
3. ***Ligaments:***
    a. **Glenohumeral ligaments** (superior, middle, inferior): thickenings of the joint capsule that strengthen the anterior aspect of capsule
    b. **Coracohumeral ligaments:** a strong band that passes from the base of the coracoid process to the anterior aspect of the greater tubercle; strengthens the capsule superiorly.
    c. **Coracoacromial ligament:** protective structure spanning from acromion to coracoid process of scapula (inferior aspect of these bony structures and ligament termed coracoacromial arch); prevents superior displacement.
    d. **Transverse humeral ligament:** runs from greater tubercle to lesser tubercle over intertubercular sulcus/groove that provides passage for long head of biceps brachii tendon

Clinical Note: The **coracoacromial arch** is so strong, extremely forceful superior thrust would sooner fracture the humerus or clavicle before fracturing the arch.

- ***Movements:*** most mobility of all joints in the body; review muscles from shoulder lecture.
- ***Blood supply:*** **anterior** and **posterior circumflex humeral a., suprascapular a.**
- ***Innervation:*** **Hilton's Law**: suprascapular n., axillary n., lateral/medial pectoral n., musculocutaneous n., upper/lower subscapular n., thoracodorsal n., radial n.

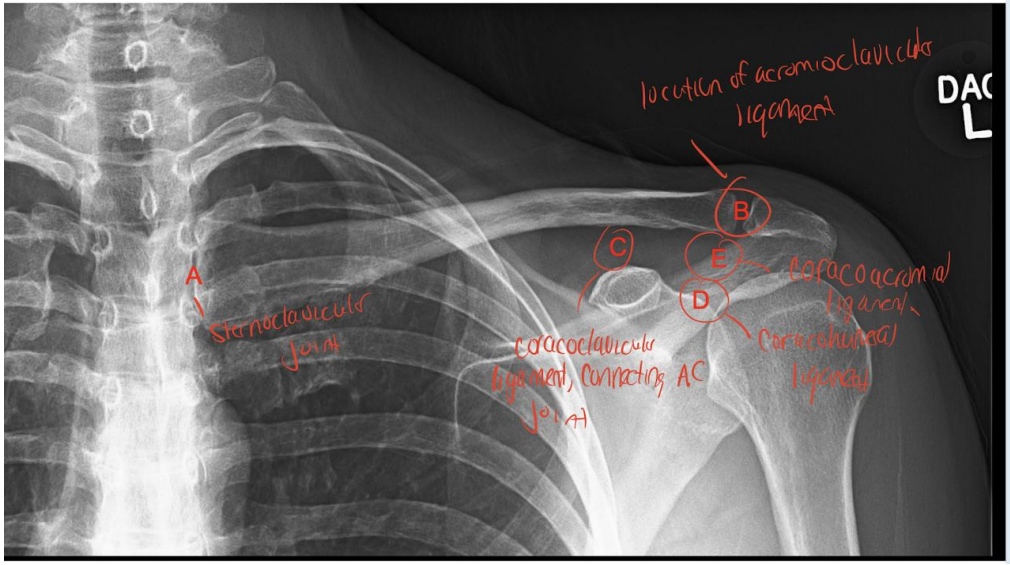
Clinical Note: dislocation of the glenohumeral joint typically happens **inferiorly** and **anteriorly (**due to the coraco-acromial arch superiorly and the **weakness of the joint capsule** antero inferiorly); the **axillary nerve** is at risk during typical dislocation.

- Treatment regimen usually includes immobilization and therapy to strengthen the rotator cuff muscles.

Clinical Note: **Shoulder Impingement/Instability**. **Shoulder Impingement Syndrome** is caused when the tendon of the **supraspinatus** becomes impinged in the subacromial space leading to tendonitis of the tendon.

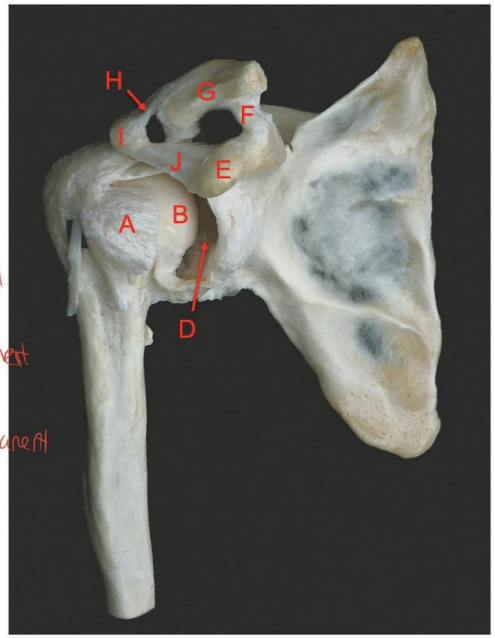
- Chronic overuse of the glenohumeral joint may stretch the static stabilizers of the shoulder causing mild shoulder instability.
- Instability puts increased stress on the rotator cuff to contain the humerus in the center of the glenoid fossa. Rotator cuff muscles become weakened leading to a subluxation of the glenohumeral joint resulting in a secondary **impingement** of supraspinatus in the subacromial space.

Knowledge Check



Knowledge Check

(A) joint capsule
(B) head of humerus
(D) glenoid cavity
(E) coracoid process
(F) corticoclavicle ligament
(G) clavicle
(H) acromioclavicular ligament
(I) acromium
(J) coracoacromial ligament



Approach to Acromioclavicular Joint Problems & Approach to Glenohumeral Joint Problems (Dr. Faber)

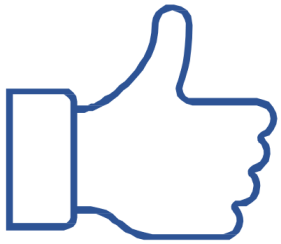
Acromioclavicular Joint Problems

Kenneth J. Faber MD, MHPE, FRCSC
Professor of Surgery
Roth McFarlane Hand and Upper Limb Centre
Western University
London, ON



AC JOINT PROBLEMS

- Are common
- And easy to sort out!!



1. AC Joint Osteoarthritis:
   - Wear and tear
2. AC Joint Instability: "Separation"
   - Injury

TREATMENT

- I: Education, sling, analgesics, local therapy (ice) and early exercises
- II: Same
- III: Same

- IV-V: Surgery



SUMMARY

- AC Joint problems are common
- Acute injury: AC Instability
- Gradual onset: Osteoarthritis
- Treatment:
  - Education
  - Activity modification
  - Local treatment: ice, injections
  - Analgesics
  - Surgery



COMMON SHOULDER DISORDERS: ARTHRITIS



Kenneth Faber MD, MHPE, FRCSC
Professor of Surgery
University of Western Ontario

ROTH | McFARLANE
HAND & UPPER LIMB CENTRE
ST. JOSEPH'S HEALTH CARE LONDON

Objectives

- Describe the typical features of shoulder arthritis:

- Noninflammatory
  - Osteoarthritis / posttraumatic
- Inflammatory arthritis:
  - RA / Psoriatic / Crystalline

- Describe treatment options

Shoulder Arthritis

- Disorder of progressive:
  - Pain
  - Reduced motion
  - Loss of function



Clinical Features - Noninflammatory

- Deep aching shoulder discomfort
- Reduced active AND passive motion
- Rotator Cuff strength usually intact



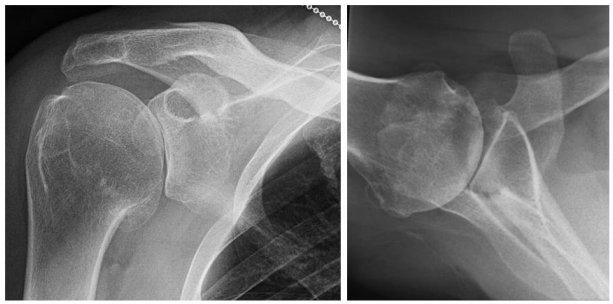
Clinical Features - Inflammatory

- Deep aching shoulder discomfort
- Active motion reduced
- Passive motion often greater than active motion
- Rotator Cuff strength usually poor



XR Features: Noninflammatory

- Asymmetric joint space
- Osteophytes
- Cysts
- Subchondral sclerosis



XR Features: Inflammatory

- Symmetric joint space
- Bone deformity
- Erosions
- Osteopenia





Treatment Options

- Education
- Activity modification
- Medications
- PT - Exercise program
- Surgery:
  - Debridement
  - Replacement



Debridement

- Reserved for young patients with arthritis
- Not curative - relieves symptoms of “catching”



Outcomes



- 85% good or excellent
- 90% survivorship; 10 yrs
- Complications:
  - Infection, loosening, instability, fracture, progressive rotator cuff disease

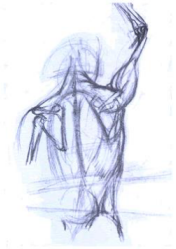
Summary

- Two clinical patterns:
  - Noninflammatory
  - Inflammatory
- Treatment:
  - Nonoperative
  - Operative
- Joint replacement effective!



SHOULDER INSTABILITY

Kenneth Faber MD, MHPE, FRCSC
Professor of Surgery
University of Western Ontario



ROTH | McFARLANE
HAND & UPPER LIMB CENTRE
ST. JOSEPH'S HEALTH CARE LONDON

OBJECTIVES

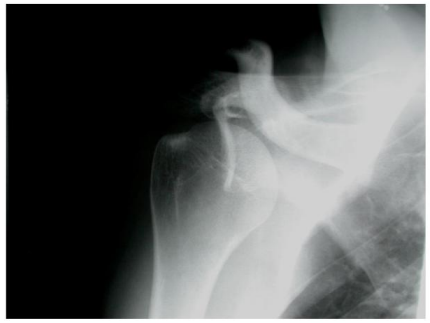
- Understand anatomic factors
- Be able to define pathology
- Understand treatment options:

1. Traumatic Instability
2. Atraumatic Instability

SHOULDER INSTABILITY

Definition:

– The inability to maintain the humeral head centered within the glenoid



Shoulder Stability

anatomic factors to consider

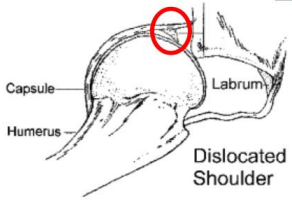
- **Ligaments** } most important
- **Labrum**
- Bones
- RC / Muscle Forces
- Nerves



Glenoid Labrum

- Glenoid labrum
  - Increased surface area
  - Increased depth
  - Bumper effect





Ligament Laxity

- Multi-directional Instability (MDI)
- ? Connective tissue disorder
- ? Poor motor control



What is the diagnosis?

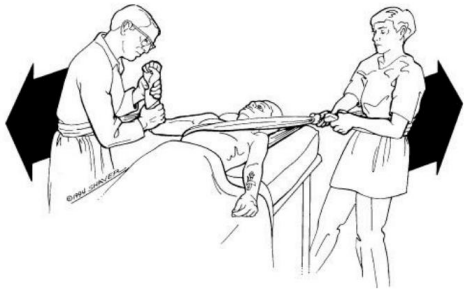
- Onset
  - traumatic, atraumatic
- Direction
  - anterior, posterior, inferior, combination
- Degree
  - subluxation vs. dislocation

OVERVIEW

TRAUMATIC	ATRAUMATIC
– Unidirectional • Anterior>>posterior – "Torn loose" – Surgery often	– Multidirectional – "Born loose" – Primary treatment: • Physiotherapy • Muscle Control – Surgery uncommon

REDUCTION METHOD

- Many described
- Principles:
  - Safety first!
    - Monitors
  - Analgesia
  - Manipulation to correct deformity
  - Re-examine after
    - Check neurovascular status



Traumatic Instability

treatment options

- Surgery:
  - Young
  - Recurrent dislocators
- Labral repair



ATRAUMATIC INSTABILITY



- AMBRI
  - Atraumatic
  - Multidirectional
  - Bilateral
  - Motor control is poor
- Treatment:
  - Rehab
  - Inferior capsular shift

ATRAUMATIC INSTABILITY

Treatment Options

- Inferior capsule shift
  - – open
  - – arthroscopic



Summary: Instability

- Anatomy: Factors that contribute to stability
- Differentiate acute / traumatic from atraumatic instability
- Treatment: nonoperative and operative

ADHESIVE CAPSULITIS



Kenneth J. Faber MD MHPE FRCSC
Professor of Surgery
Western University

ROTH | McFARLANE
HAND & UPPER LIMB CENTRE
ST. JOSEPH'S HEALTH CARE LONDON

Objectives

- Differentiate causes of shoulder stiffness
- Describe the stages
- Develop an algorithm for treatment

Background



- 2% of adults
- 20% of patients with diabetes
- Onset 40's and 50's
- Pathophysiology unknown

Shoulder Stiffness

1. Intraarticular
   - Arthritis
   - Posttraumatic joint deformity
2. Extraarticular
   - Capsular
   - Muscular
   - Heterotopic bone
3. Combined



What is it?

- Spontaneous onset
- Global restriction in motion
- Active and passive
  - *"A condition of uncertain etiology characterized by a restriction of active and passive motion that occurs in the absence a known intrinsic shoulder disorder"*

ASES

Associated Conditions

- **Diabetes**
- Most often idiopathic
- Cervical disc
- Thyroid disease
- Cardiac disease



Case: 47 Yr Old Male

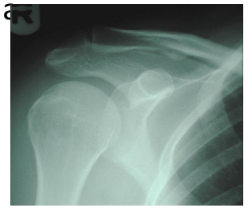
- 4-month history of Lt shoulder pain
- Progressive loss of motion
- Diabetic



Differential Diagnosis

1. Arthritis
2. Post trauma deformity

- BUT…
  - Xrays normal
  - No trauma



Treatment

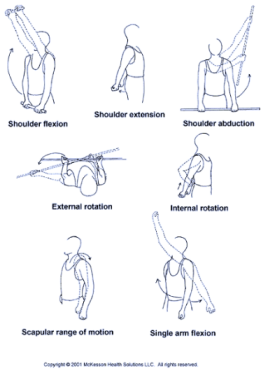


- Education
- Pain management
- Physiotherapy
- Injections:
  - Steroids
  - Hydroplasty – distention
- Surgery:
  - Manipulation
  - Arthroscopic capsulectomy

Physiotherapy

- Controversial
- No quality studies demonstrate a significant effect with PT

AlSubheen 2019



Injections: Steroids



- Intra-articular steroid superior to PT
- No Difference @ 1 yr

Injections: Hydrodistention



- Briser: fr. verb “to break”
- Inject liquid under pressure.

Surgery

More rapid return of motion

Similar to all other treatments @ 2 years

Summary

- A common disorder
- Hallmarks: restricted active and passive motion
- Common in patients with diabetes
- Usually self-limited